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JW-FD_V2: A 15-Year Multimodal Dataset for Solar Flare Forecasting

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Abstract

Solar flares drive severe space weather hazards, and forecasting their occurrence remains a central challenge for both heliophysics and operational space weather services. Data driven methods require long horizon datasets in which images, magnetic features, and flare labels are coregistered in space and time. We present **JW-FD** (JinWu-Flare Dataset), a 15 year multimodal release spanning 1 January 2011 through 31 December 2025, constructed from SDO/HMI line of sight magnetograms, NOAA Solar Region Summary reports, and NOAA X-ray flare event lists. The dataset comprises 3,083 independent active regions and 2,003,969 coregistered magnetogram crops, together with FITS, PNG, CSV, and MP4 modalities. Each sample provides 29 magnetic features linked to configurable flare labels under a strict pre-eruption window spanning seven forecast horizons and four GOES intensity thresholds. An 8:1:1 split at the active region level, obtained by stratified rejection sampling (seed 3970), keeps each active region in a single partition to prevent temporal leakage and balances rare M- and X-class positives between validation and test. PNG branches are released at ten magnetic saturation thresholds. A Swin ablation over all ten clips finds $\geq$C1.0 skill nearly insensitive to B_{th}, whereas $\geq$M1.0, $\geq$M5.0, and $\geq$X1.0 lock different clips. We therefore adopt $B_{\mathrm{th}} = 800$ G as a compromise PNG working point rather than the optimum of any one task, and report four-model 24 h baselines at that clip. Skill remains modest as the class becomes rarer. The open source construction pipeline is available at https://github.com/Xiaoxuan-1/JW-FD, and the released dataset (version v2) at https://doi.org/10.57760/sciencedb.019mg.

Keywords: solar flare forecasting; magnetograms; multimodal dataset; machine learning

1. Introduction

Solar flares are among the most energetic manifestations of solar magnetic activity. Major (M- and X-class) flares can disrupt satellites, communications, and power grids, making reliable flare forecasting a priority for both fundamental solar physics and operational space weather services [1–3].

Photospheric line of sight (LoS) magnetograms encode signatures of magnetic complexity (gradient concentrations, neutral line topology, and flux imbalance) that precede many eruptive events [4,5]. Quantitative features derived from magnetograms have long supported statistical and machine learning (ML) flare forecasting, and recent work extends this paradigm to convolutional neural networks (CNNs), vision transformers, and multimodal models that consume both images and structured parameters [6,7]. These methods

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require datasets in which images, physical features, and flare labels are coregistered in space and time, under a partitioning scheme that prevents temporal leakage across splits.

Several public datasets now support data driven flare forecasting, but they differ in temporal span, modality coverage, and evaluation design. Boucheron et al. released an AR LoS magnetogram image corpus covering 2010–2018 [8]. Angryk et al. released SWAN-SF, a multivariate time series corpus of Space weather HMI Active Region Patch (SHARP) parameters over a similar interval [9,10]. Nishizuka et al. assembled feature databases associated with Deep Flare Net (DeFN) [11,12], and related work continues to organize SHARP parameter sets for operational forecasting [13,14]. Complementary studies have assembled magnetogram or multiwavelength datasets for CNNs, long short-term memory (LSTM) networks, and related models [15–18], yet many remain study-specific and are not publicly redistributed as reusable, long-horizon, multimodal corpora [1,19,20].

More recently, SuryaBench provides a large multitask heliophysics benchmark for foundation model research, with solar flare forecasting as one of several applications [21]. Its flare subset uses full disk SDO context with a fixed 24 h labeling window and a calendar based train/test partition, rather than NOAA AR crops with configurable multi horizon labels and an AR level split. Relative to Boucheron et al. [8], SWAN-SF [9], and the SuryaBench flare task [21], the JinWu-Flare Dataset (JW-FD) supplies a long horizon, AR centered multimodal release in which images, magnetic features, flare labels, and evolution videos share a common index under a leakage aware AR level partition [3].

JW-FD is designed to meet this need. This paper presents the dataset, its design rationale, construction pipeline, and recommended usage. Our principal contributions are as follows:

1. A **15 year multimodal release** (2011–2025) covering solar Cycles 24–25 with 2.00 million active region magnetogram crops, supporting studies of cycle dependent flare statistics and rare X class events.
2. **Coregistered images, magnetic features, flare labels, and evolution videos**, with 66 dimensional records that link 29 magnetic features to configurable labels defined before eruption.
3. **Configurable labels** over seven forecast horizons (1–72 h) and four GOES intensity thresholds (C1.0–X1.0), defined on a strict pre-eruption window.
4. **An 8:1:1 training, validation, and test split.** The data are divided into training, validation, and test sets in an 8:1:1 ratio, with all observations from the same active region assigned to a single partition. This prevents information from the same region at different times from leaking between the datasets. The assignment is a stratified rejection sample (seed 3970) that equalizes rare-class positives between validation and test at all seven forecast horizons.

2. Methods

JW-FD is constructed with a six step automated pipeline that ingests Solar Region Summary (SRS) reports and NOAA Events X-ray (XRA) flare lists from the National Oceanic and Atmospheric Administration/Space Weather Prediction Center (NOAA/SWPC), together with Solar Dynamics Observatory (SDO)/HMI [22,23] full disk LoS magnetograms. The pipeline exports Flexible Image Transport System (FITS) crops, Portable Network Graphics (PNG) images, comma separated values (CSV) tables, and MPEG-4 (MP4) videos. Figure 1 summarizes the workflow; Table 1 lists the inputs and outputs of each step.

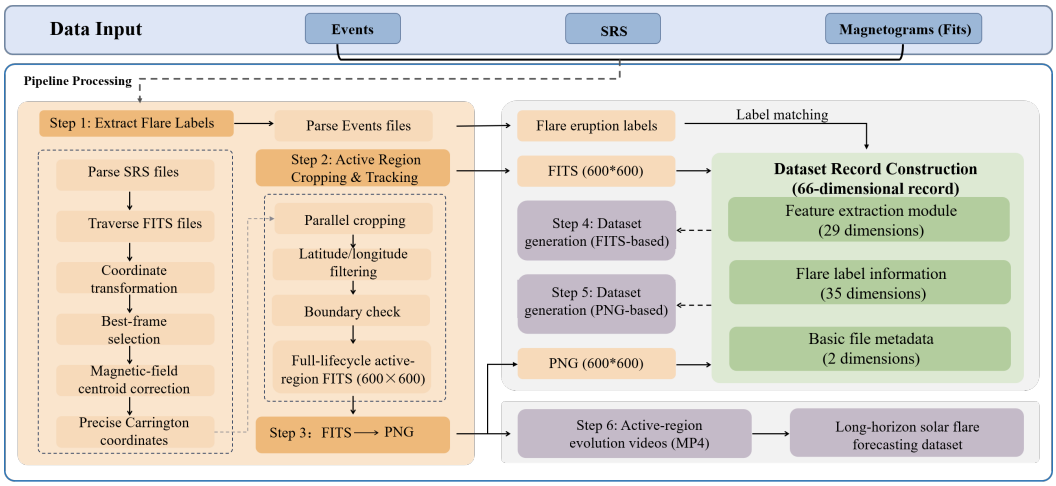


Figure 1. JW-FD dataset construction workflow. Step 1 extracts NOAA X-ray flare events. Step 2 crops and tracks NOAA active regions on SDO/HMI full-disk line-of-sight magnetograms. Step 3 converts FITS crops to 8-bit PNG images at ten magnetic saturation thresholds. Steps 4 and 5 extract the same 29-dimensional feature set and share label-matching logic on the FITS and PNG branches, respectively. Step 6 encodes per-active-region PNG sequences as evolution videos.

Table 1. Pipeline step inputs and outputs. Here B_{th} denotes the magnetic saturation threshold (in Gauss) applied when converting FITS magnetograms to 8-bit PNG images; the release includes ten branches from 200 to 2000 G in steps of 200 G.

Step	Script	Input	Output
1	`step1_extract_flare_labels`	NOAA Events `.txt` (XRA)	`flare_labels.csv`
2	`step2_crop_active_regions`	SRS + HMI FITS (4096^2)	`fits_600/{AR}/`
3	`step3_convert_to_png`	Step 2 FITS	`png_600_Th{`B_{th}`}/`
4	`step4_generate_dataset_fits`	FITS + labels	FITS branch CSV
5	`step5_generate_dataset_png`	PNG + labels	PNG branch CSV
6	`step6_make_movie`	PNG sequences	`movies/AR{num}_evolution.mp4`

2.1. Data Sources

SDO/HMI provides full disk LoS magnetograms at 4096×4096 pixels (∼1.5″ per pixel) with a cadence of approximately 12 minutes. The source frames are not projection-corrected: they remain in the observer plane, and B_{LoS} is not converted to an approximate radial field. **NOAA/SWPC SRS** reports list daily NOAA AR numbers and heliographic locations. **NOAA Events** files record GOES soft X-ray flare begin, maximum, and end times, together with GOES class strings (C, M, or X followed by a numeric subclass).

2.2. Design Choices

Four design choices distinguish JW-FD from SHARP based patch datasets and earlier AR image datasets [5,8].

Long temporal coverage. Fifteen calendar years (2011–2025) span the declining phase of Solar Cycle 24, the 2018–2020 minimum, and the rising phase of Cycle 25. This span supports cycle dependent statistics and supplies enough rare M and X class events for threshold specific studies.

Multimodal coregistration. Every sample is indexed by NOAA AR number and Coordinated Universal Time (UTC) timestamp across FITS magnetograms (Gauss scale physics), PNG frames (vision model input), 29 dimensional magnetic feature vectors, configurable flare labels, and per AR MP4 evolution videos. Users may train end to end vision models on PNG or MP4 inputs, parameter based classifiers on extracted features, or multimodal models that fuse imagery with structured metadata [6].

Configurable pre-eruption labels. Binary and class labels are defined for seven forecast horizons ($h \in \{1, 3, 6, 12, 24, 48, 72\}$ hours) and four GOES thresholds ($\tau \in \{\text{C1.0}, \text{M1.0}, \text{M5.0}, \text{X1.0}\}$). Positive labels are assigned only within the half open interval $[\text{Begin} - h, \text{Begin})$, so frames at or after eruption onset are not treated as positive (Figure 2).

AR level evaluation split. Consecutive HMI frames of the same AR exhibit strong temporal autocorrelation, so assigning different timestamps of one AR to both training and test sets constitutes information leakage [3,20]. JW-FD therefore adopts an 8:1:1 split at the AR level (Section 4.1), obtained by stratified rejection sampling so that rare-class positives are balanced between validation and test.

Additional practical choices include: (i) 600×600 pixel crops (∼300″ field of view); (ii) NOAA AR centered directory organization rather than HMI Active Region Patch (HARP)/SHARP patch identifiers, matching operational numbering; (iii) a disk center filter ($|\text{longitude}| \leq 60°$) that reduces the most severe limb effects but does not apply a geometric projection correction; and (iv) dual FITS and PNG CSV branches so that physics faithful (Gauss scale) and vision ready (normalized intensity) feature extraction share identical label logic.

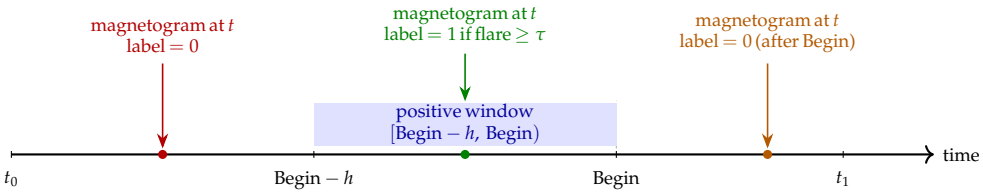


Figure 2. Pre-eruption labeling window for horizon h. A magnetogram observed at time t receives a positive binary label for threshold τ only if $t \in [\text{Begin} - h, \text{Begin})$ and the matched GOES class meets or exceeds τ. Observations at or after Begin (dashed line) are excluded from the positive class.

2.3. Active Region Cropping and Tracking

Tracking begins from the SRS Stonyhurst location of each NOAA AR at its first available magnetogram time, which is converted to a Carrington anchor. That anchor is then carried across subsequent epochs so that the same heliographic region remains centered as the Sun rotates. Local refinement masks pixels stronger than 100 G inside an 80 pixel neighborhood of the projected center and updates the center to their mass centroid. From each full disk FITS frame we extract a 600 × 600 crop and discard ARs with $|\text{longitude}| > 60°$.

2.4. FITS to PNG Conversion

Cropped FITS magnetograms are converted to 8 bit PNG images for subsequent vision based analysis. Each crop is mirrored left to right to follow the conventional north up, east left display orientation. Magnetic field values are then clipped at a chosen saturation threshold and linearly mapped to the integer range $[0, 255]$:

$$B' = \text{clip}(B, -B_{\text{th}}, B_{\text{th}}), \quad I = \left\lfloor \frac{B' + B_{\text{th}}}{2B_{\text{th}}} \times 255 \right\rfloor, \tag{1}$$

Invalid (NaN) samples are filled with 0 G prior to clipping, so that 0 G occupies grey level 127. We publish PNG trees for ten saturation levels, $B_{\text{th}} = 200$–2000 G in steps of 200 G, each with matched 29-dimensional features and flare labels; Section 3.2 discusses how this choice affects downstream models.

2.5. Feature Extraction and Label Matching

Steps 4 and 5 generate CSV datasets from FITS and PNG inputs. Both branches extract the same 29 magnetic features (Table 2) and apply the same label-matching logic, but they operate on different physical domains. FITS features are computed on Gauss-scale magnetograms with no grey-level offset; invalid (NaN) pixels are replaced by 0 G before extraction. PNG features are computed on 8-bit intensity images after subtracting a 127-level zero point, which is the uint8 bin that contains 0 G under a fixed $\pm B_{\mathrm{th}}$ stretch (Equation 1). The 29 features follow the snapshot (non-temporal) subset of the photospheric proxies introduced by Al-Ghraibah et al. [24] and subsequently used in a multivariate time-series setting by Boucheron et al. [4]. Al-Ghraibah et al. extracted flux, gradient, neutral-line, flux-evolution, and wavelet descriptors from line-of-sight magnetograms and classified whether an active region would produce a flare of GOES class $\geq$C1.0. JW-FD retains the 7 gradient, 13 neutral-line, 5 wavelet, and 4 flux descriptors (29 in total) as per-frame attributes; the 9 flux-evolution descriptors of Al-Ghraibah et al. are not precomputed, because the released image and video sequences already allow users to form temporal differences if desired. Operationally, (i) Sobel filters yield a gradient-magnitude map from which we store the mean, standard deviation, median, extrema, skewness, and kurtosis; (ii) a 10×10 boxcar mean (implemented as a uniform filter, with near-zero window means reset to 0 G to avoid spurious contours from floating-point residue) and zero-Gauss contours define polarity inversion lines, which are then weighted by gradient magnitude to obtain length, fragment count, gradient-weighted length, and statistics of curvature and bending energy; (iii) a five-level Haar decomposition yields orientation-summed sub-band energies L1–L5; and (iv) total positive, negative, signed, and unsigned flux are obtained by summing pixel values over the corresponding polarity masks. These quantities are physically interpretable proxies of magnetic complexity and free-energy storage, and the open-source pipeline regenerates the same numbers from the released FITS crops.

Figure 3 illustrates the four feature families on NOAA AR 11158 at 17:00 UT on 13 February 2011. Panel (a) shows the line-of-sight magnetogram from which the four flux integrals are computed. Panel (b) is the Sobel gradient-magnitude map that supplies the seven gradient statistics. Panel (c) shows the neutral-line descriptors: thin lime dashes are all zero-Gauss contours after 10×10 smoothing (the noisy inversion line), and thin magenta markers mark only the high-gradient subset of the gradient-weighted neutral line (top 10% of $|\nabla B|$ on the contour), with marker size scaled by $|\nabla B|$. Panel (d) displays the five-level Haar coefficient array whose orientation-summed sub-band energies are stored as L1–L5.

Table 2. Twenty-Nine Magnetic Features and Their CSV Column Names

Category	Feature names
Gradient (7)	Gradient mean; Gradient std; Gradient median; Gradient min; Gradient max; Gradient skewness; Gradient kurtosis
Neutral line (13)	NL length; NL no. fragments; NL gradient-weighted length; NL curvature mean/std/median/min/max; NL bending energy mean/std/median/min/max
Wavelet (5)	Wavelet Energy L1; Wavelet Energy L2; Wavelet Energy L3; Wavelet Energy L4; Wavelet Energy L5
Flux (4)	Total positive flux; Total negative flux; Total signed flux; Total unsigned flux

Relative to image-only AR corpora [8], SHARP parameter time series [9], and full-disk flare benchmarks [21], JW-FD is distinctive here in that the 29 magnetic features, configurable flare labels, and FITS/PNG/MP4 products share a common AR-and-time index and can be regenerated from the released crops by the open-source pipeline.

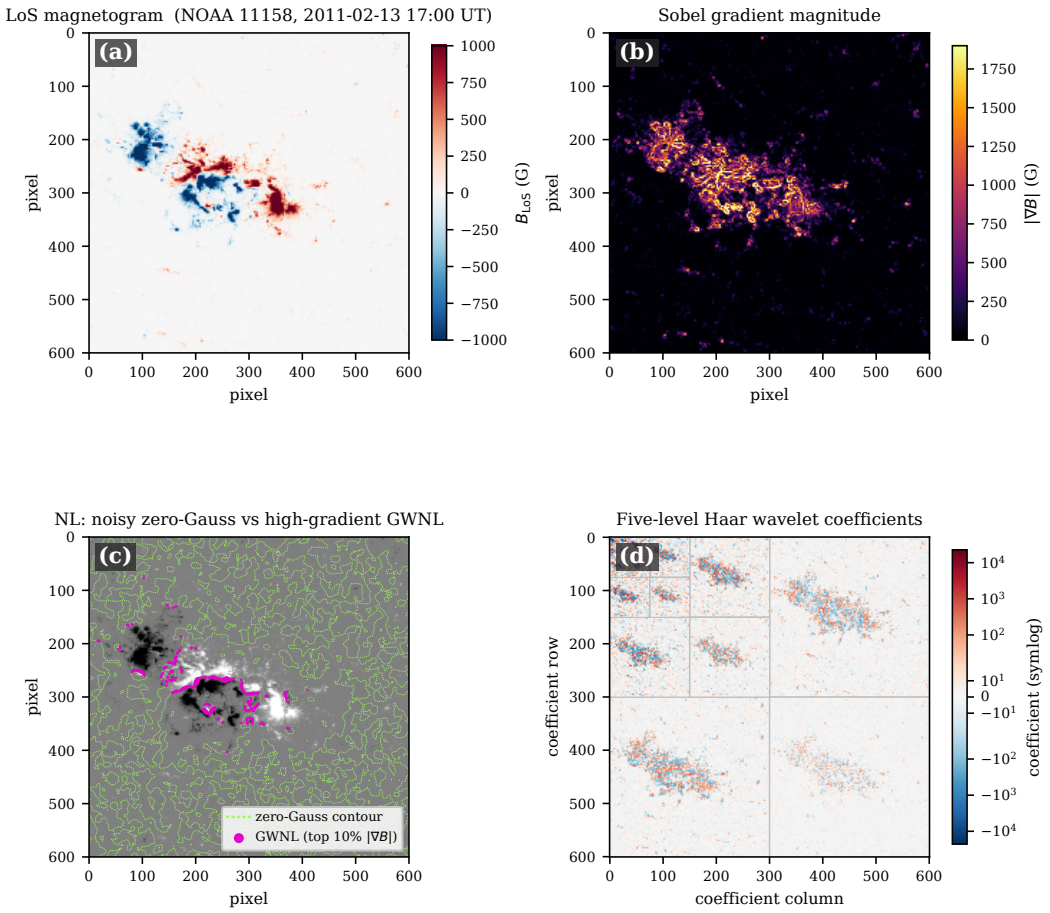


Figure 3. Feature extraction on NOAA AR 11158 at 17:00 UT on 13 February 2011 (FITS crop, 600×600 pixels), following Al-Ghraibah et al. [24]. (a) Line-of-sight magnetogram. (b) Sobel gradient magnitude. (c) Neutral-line contours and the gradient-weighted subset. (d) Five-level Haar wavelet coefficients.

Each CSV row corresponds to one magnetogram time stamp. We define seven forecast horizons $h \in \{1, 3, 6, 12, 24, 48, 72\}$ hours and four GOES thresholds $\tau \in \{\text{C1.0}, \text{M1.0}, \text{M5.0}, \text{X1.0}\}$. The binary column `flare_label_{`$\tau$`}_{`$h$`hr}` equals 1 when at least one flare in the same AR has GOES class $\geq \tau$ and a begin time Begin in the half open window

$$\text{Begin} - h \;\leq\; t \;<\; \text{Begin} \tag{2}$$

and equals 0 otherwise. Separately, `flare_class_{`h`hr}` stores the strongest GOES class among flares that fall in the same window for horizon h, and equals `0` when no such flare occurs.

2.6. Evolution Videos

For each AR directory, PNG frames are sorted chronologically and encoded as H.264 MP4 at 60 frames per second. Each AR produces one `AR{num}_evolution.mp4` file in the `movies/` directory. The 60 fps setting is a playback rate for visualization, not an observational cadence: consecutive frames remain the native HMI LoS crops sampled at approximately 12 minutes. At this encoding, one second of video spans about 12 hours of solar time, so the MP4 files provide a time compressed overview of AR evolution for inspection and qualitative analysis rather than a high temporal resolution data stream.

2.7. Quality Control and Expert Verification

Quality control is enforced throughout the pipeline. Flare lists are first screened to remove records with invalid AR identifiers, missing peak times, or unparseable GOES class strings. During active region extraction, crops are retained only when their centers remain on the usable disk and the AR longitude lies within the prescribed limit. Subsequent conversion and feature extraction stages record truncated or unreadable inputs in step specific logs, skip empty files, and replace NaN values encountered during PNG normalization and FITS feature extraction (filled with 0 G). These automated checks operate together with the feature pipeline in Section 2.5 so that invalid flare records, unusable disk positions, and truncated or NaN inputs are excluded before the 29 magnetic parameters and flare labels are written.

In addition to these automated checks, solar physicists inspected a randomly selected 5% subset of the processed samples. Inspection criteria included crop centering, completeness of the field of view, and consistency of M/X class labels with the intended windows before eruption. Samples affected by telemetry gaps or uncommon processing failures were excluded or regenerated.

3. Results

3.1. Dataset Records and Statistics

Table 3 summarizes the scale of the JW-FD release. Here each magnetogram sample is one 600×600 active region crop; the remaining rows inventory the accompanying modalities, flare event records, and evaluation split.

Table 3. Summary statistics of the JW-FD release (2011–2025).

Quantity	Specification
Temporal coverage	1 January 2011 – 31 December 2025 (15 yr)
Active regions	3,083 independent NOAA ARs
Magnetogram samples	2,003,969
Spatial resolution	600×600 pixels per AR crop
Released modalities	FITS, PNG, CSV, and MP4 (3,083 videos)
Flare event records	18,986 NOAA XRA events (`flare_labels.csv`)
Evaluation split	AR level 8:1:1 (training / validation / test)
Data sources	SDO/HMI LoS; NOAA/SWPC SRS; NOAA Events (XRA)

The 18,986 flare event records are the NOAA XRA events in 2011–2025 that carry an AR number (`flare_labels.csv` contains 21,506 rows; the other 2,520 are events from 2010 or events without an AR number, and cannot be assigned to samples).

Annual sample totals are strongly modulated by the solar cycle (Figure 4). Counts peak near the Cycle 24 maximum (2013–2014) and again during the Cycle 25 rise (2023–2025), whereas the 2018–2020 activity minimum yields far fewer NOAA numbered ARs and therefore fewer retained samples.

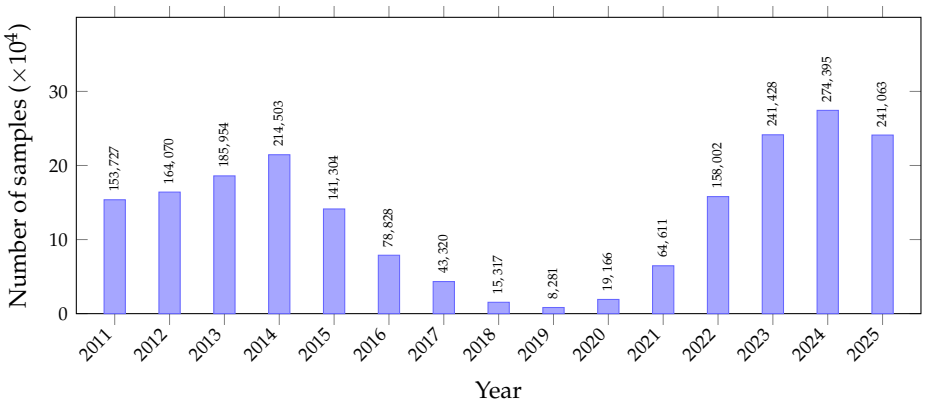


Figure 4. Annual distribution of magnetogram samples in JW-FD. Sample counts track solar activity, with peaks near the Cycle 24 maximum (2013–2014) and the Cycle 25 rise (2023–2025), and a clear minimum in 2018–2020.

Processed products share a common release root and are partitioned by modality (Table 4). Magnetogram and label directories are indexed by NOAA AR number; PNG branches further separate data by the saturation threshold B_{th}.

Table 4. Released data modalities and directory layout.

Modality	Content	Directory layout
FITS	LoS magnetogram in Gauss; 600 × 600 crop	`fits_600/{ARnum}/...`
PNG	8 bit grayscale; saturation B_{th} encoded in path	`png_600_Th{B_th}/{ARnum}/...`
CSV	66 dimensional record per sample; FITS and PNG branches	`label/{fits|png}/Th{B_th}/...`
MP4	Per AR time compressed evolution (60 fps playback; ∼12 min source cadence)	`movies/AR{num}_evolution.mp4`
Labels	NOAA XRA flare event list	`flare_labels.csv`

Each CSV row indexes one sample under a fixed 66 dimensional schema (Table 5). The same record can therefore support tabular learning, vision models, or multimodal fusion without reconstructing labels externally.

Table 5. Composition of each 66 dimensional CSV record.

Component	N	Description
Magnetic features	29	Gradient, neutral line, wavelet, and flux descriptors (Table 2)
Binary flare labels	28	Seven forecast horizons × four GOES thresholds
Class labels	7	Strongest GOES class within each forecast horizon
File metadata	2	`image_filename` and `image_path`
Total	**66**	

For evaluation, JW-FD provides an AR-level 8:1:1 split (Table 6). The assignment is a stratified rejection sample of active regions. Each NOAA region is placed in one flare-productivity stratum: multiple X-class, single X-class, M5 but not X, M1 but not M5, C-class only, or flare quiet. A stratum is divided into three tertiles by magnetogram-frame count when each tertile still contains at least 20 regions; smaller strata remain a single cell. Within each cell the regions are shuffled and cut 8:1:1, and every timestamp of a region follows that region into one partition. Candidate seeds are tried in order from 62, and a seed is discarded unless every gate passes. Validation and test must each contain at least five X-class and five M5-class positive regions. Separately for the ≥C1.0, ≥M1.0, ≥M5.0, and ≥X1.0 grades, the validation-to-test ratio of 24 h positive frames must lie in $[0.5, 2]$, so that validation and test

remain balanced. The flare-productivity strata equalize how many regions of each class fall in validation and test, but regions within one stratum can still differ in size and magnetic complexity. Each region is therefore summarized by two frame-wise medians, taken over all of its magnetograms: total unsigned flux, which measures how much magnetic flux the crop contains, and neutral-line length, which measures how extended the polarity boundary is. Cohen d is the difference between the validation and test means of each median, divided by their pooled standard deviation, and is computed over every region in the two partitions. A seed is rejected when $|d| \geq 0.30$ for either quantity, that is, when one hold-out set is shifted by 0.3 standard deviations or more toward larger or more complex regions. Because the decision threshold is selected on the validation set, this gate keeps that choice from being tuned on a magnetically easier or harder population than the test set. Seed 3970 is the first seed that meets all of these gates. All samples from one AR are assigned to a single subset; sample fractions may depart slightly from 8:1:1 because ARs contribute unequal numbers of magnetograms. Validation and test each contain 307 regions. The 24 h positive regions are seven X-class and 16 M5-class in validation, and seven X-class and 14 M5-class in test. Table 7 lists the training, validation, and test positive-frame counts at all seven forecast horizons.

Table 6. Active-region-level partitioning from stratified rejection sampling (seed 3970; nominal ratio 8:1:1).

Subset	Samples (N)	Fraction	ARs (N)	Fraction
Training	1,604,907	80.09%	2,469	80.08%
Validation	202,019	10.08%	307	9.96%
Test	197,043	9.83%	307	9.96%
Total	**2,003,969**	**100%**	**3,083**	**100%**

Table 7. Positive-frame counts in the training, validation, and test partitions under seed 3970, for all seven forecast horizons and four GOES thresholds.

Horizon	Subset	≥C1.0	≥M1.0	≥M5.0	≥X1.0
1 h	Train	43,591	5,776	874	325
	Val	6,016	942	161	50
	Test	5,487	764	95	45
3 h	Train	109,206	15,958	2,575	955
	Val	15,303	2,629	471	139
	Test	13,366	2,122	284	135
6 h	Train	179,277	29,003	4,996	1,886
	Val	25,105	4,725	907	272
	Test	21,644	3,874	569	269
12 h	Train	275,088	49,732	9,607	3,683
	Val	38,414	8,033	1,603	512
	Test	32,877	6,669	1,069	507
24 h	Train	399,097	79,466	16,797	6,621
	Val	54,146	12,929	2,811	939
	Test	47,794	11,231	2,064	968
48 h	Train	546,922	119,912	28,396	11,744
	Val	72,611	19,684	4,776	1,807
	Test	65,017	17,853	4,045	1,898
72 h	Train	634,729	151,606	37,904	16,171
	Val	83,171	24,132	6,297	2,433
	Test	74,990	21,942	5,602	2,542

3.2. Magnetic Saturation Threshold Ablation

Converting Gauss scale FITS magnetograms to 8 bit PNG requires a saturation level B_{th} in Equation (1). This choice affects visual contrast, feature statistics on PNG derived branches, and model input distributions. JW-FD therefore releases PNG branches at ten saturation levels, $B_{\mathrm{th}} = 200$–2000 G in steps of 200 G, each accompanied by 29-dimensional magnetic features and the same configurable flare labels.

To provide an initial reference point, we ran a controlled ablation over all ten values with a Swin-B vision classifier on the PNG branch, using binary flare grades $\geq$C1.0, $\geq$M1.0, $\geq$M5.0, and $\geq$X1.0 and the seed 3970 AR-level assignment. Thresholds are compared at a fixed decision point of 0.5 on the validation set (val TSS@0.5), so that clip differences are not confounded by per-threshold tuning. Table 8 reports that validation score together with the corresponding test TSS@0.5, ROC-AUC, and PR-AUC. Across the ten clips, val TSS@0.5 for $\geq$C1.0 spans only 0.5130–0.5371, so $\geq$C1.0 skill is nearly insensitive to B_{th}.

Unlike $\geq$C1.0, the rarer-class tasks are more sensitive to B_{th} and do not share a val TSS@0.5 maximum: the lock is 400 G for $\geq$M1.0 (range 0.6030–0.6799), 2000 G for $\geq$M5.0 (0.7300–0.7704), and 600 G for $\geq$X1.0 (0.6466–0.7548). A monotonic trend in which stronger flares simply prefer higher clips is therefore not supported.

A low B_{th} saturates $|B| > B_{\mathrm{th}}$ and allocates grey levels to weak field and polarity-inversion structure; a high B_{th} preserves unsaturated umbral and penumbral field. The $\geq$C1.0 plateau is consistent with either cue being sufficient. The $\geq$M1.0 preference for 200–400 G is compatible with inversion-line contrast, but this experiment uses PNG images rather than an ablation of the 29 magnetic-feature families, so that reading remains an interpretation. On the $\geq$M5.0 validation set the high clips 800–2000 G form a cluster (800 G lies within 0.003 of 2000 G), which would fit a strong-field requirement. The $\geq$M5.0 test set does not confirm that pattern: test TSS@0.5, TSS at the validation-tuned operating point, ROC-AUC, and PR-AUC all favour mid-to-low clips (near 600 G), and 2000 G is among the weakest test scores. The $\geq$X1.0 grid is still more sample-limited and is not used to choose the working point.

The 800 G clip is first in none of the four task blocks, yet it is never among the weakest. We therefore adopt $B_{\mathrm{th}} = 800$ G as a single compromise working point for the released PNG branch, not as the optimal clip of any one task.

We emphasize that this threshold selection is a preliminary exploration limited to the Swin architecture and a snapshot 24 h setting. Because JW-FD releases PNG branches at all ten B_{th} values, users should treat the 800 G compromise as a starting point rather than a universal optimum and re-ablate when the model, horizon, or GOES threshold differs.

3.3. Baseline Forecasting Experiments

To illustrate how JW-FD can be used for both parameter-based and vision-based forecasting, we trained four baselines on the seed 3970 AR-level assignment at the compromise clip $B_{\mathrm{th}} = 800$ G: an RBF support vector machine on the 29 magnetic features, and ResNet-152, ViT-B/16, and Swin-B classifiers on the PNG crops. These experiments are usage demonstrations of the released records at that shared clip, not a further per-model threshold search. The task is binary 24 h forecasting at GOES thresholds $\geq$C1.0, $\geq$M1.0, $\geq$M5.0, and $\geq$X1.0. Training used class-balanced losses; all active regions were retained, while magnetogram frames were temporally subsampled. Testing used the full test partition at natural class prevalence. The decision threshold was selected on the validation set to maximize TSS.

Table 9 reports test-set skill. For $\geq$C1.0, the three vision models attain TSS of order 0.52–0.55 and F1 of order 0.57–0.58, while SVM is slightly lower (TSS 0.49). For $\geq$M1.0, TSS rises to about 0.61–0.70, while precision and F1 are lower because positives are rarer. The

Table 8. Forecast performance under different magnetic saturation thresholds B_{th} (Swin-B, 24 h, seed 3970) for ≥C1.0, ≥M1.0, ≥M5.0, and ≥X1.0. Clips are compared at a fixed decision threshold of 0.5. The compromise working point 800 G is shown in bold in each task block. The ≥X1.0 block is reported for completeness and is not used to select the release default.

Task	B_{th}	val TSS@0.5	test TSS@0.5	test ROC	test PR
≥C1.0	200 G	0.5130	0.5078	0.8507	0.6110
	400 G	0.5242	0.5317	0.8487	0.6078
	600 G	0.5256	0.5251	0.8566	0.6122
	800 G	**0.5298**	**0.5282**	**0.8573**	**0.6205**
	1000 G	0.5363	0.5327	0.8576	0.6207
	1200 G	0.5364	0.5266	0.8588	0.6238
	1400 G	0.5317	0.5440	0.8639	0.6239
	1600 G	0.5328	0.5326	0.8599	0.6222
	1800 G	0.5371	0.5317	0.8590	0.6161
	2000 G	0.5326	0.5196	0.8579	0.6142
≥M1.0	200 G	0.6698	0.6396	0.9128	0.3625
	400 G	0.6799	0.6439	0.9161	0.3661
	600 G	0.6404	0.6501	0.9146	0.3896
	800 G	**0.6564**	**0.6746**	**0.9178**	**0.3758**
	1000 G	0.6166	0.6058	0.9178	0.3922
	1200 G	0.6135	0.6147	0.9203	0.4065
	1400 G	0.6089	0.6230	0.9229	0.4207
	1600 G	0.6244	0.6397	0.9248	0.4250
	1800 G	0.6030	0.5951	0.9249	0.4211
	2000 G	0.6140	0.6042	0.9256	0.4189
≥M5.0	200 G	0.7344	0.7938	0.9531	0.1020
	400 G	0.7300	0.8083	0.9548	0.1165
	600 G	0.7413	0.8151	0.9544	0.1207
	800 G	**0.7675**	**0.7761**	**0.9501**	**0.1153**
	1000 G	0.7588	0.7300	0.9461	0.1118
	1200 G	0.7500	0.7248	0.9437	0.1085
	1400 G	0.7483	0.7323	0.9435	0.1066
	1600 G	0.7589	0.7316	0.9461	0.1104
	1800 G	0.7681	0.7292	0.9482	0.1077
	2000 G	0.7704	0.7145	0.9475	0.1037
≥X1.0	200 G	0.6743	0.8790	0.9602	0.0459
	400 G	0.7422	0.8896	0.9616	0.0470
	600 G	0.7548	0.8774	0.9612	0.0464
	800 G	**0.7255**	**0.8892**	**0.9609**	**0.0463**
	1000 G	0.7080	0.8852	0.9608	0.0478
	1200 G	0.6858	0.8688	0.9572	0.0448
	1400 G	0.6711	0.8285	0.9549	0.0433
	1600 G	0.6672	0.7905	0.9535	0.0426
	1800 G	0.6487	0.7394	0.9522	0.0422
	2000 G	0.6466	0.7397	0.9512	0.0419

same pattern continues at ≥M5.0 and ≥X1.0: TSS remains high (0.65–0.84 for the vision models), but precision, F1, and PR-AUC collapse. The area under the receiver operating characteristic curve (ROC-AUC) and the area under the precision–recall curve (PR-AUC) do not depend on the TSS-tuned decision threshold. ROC-AUC asks only whether flare frames receive higher scores than non-flare frames; it remains 0.82–0.86 for ≥C1.0 and 0.88–0.92 for ≥M1.0, and stays above 0.84 even for ≥X1.0. PR-AUC asks how often a high score is a true flare, and therefore tracks prevalence: it falls from 0.50–0.62 at ≥C1.0 to 0.30–0.39 at ≥M1.0, then to 0.04–0.12 at ≥M5.0 and 0.01–0.05 at ≥X1.0, in line with the FAR above 0.78 for ≥M1.0 and above 0.92 for ≥M5.0. Figure 5 shows the corresponding ROC and precision–recall curves. These C- and M-class scores are comparable in magnitude to earlier LoS-parameter studies [24]. Dedicated SHARP-based systems such as FlareCast [25]

report higher test-set skill, and two points help to place that comparison in context. First, FlareCast takes as input sequences of SHARP radial-field (B_r) magnetograms (six frames spanning 6 h), rather than line-of-sight PNG crops, and it first regresses the maximum X-ray flux before assigning a flare class. Second, the 24 h test-set TSS of 0.891 for ≥C is accompanied by lower values for ≥M (0.792) and ≥X (0.831); in an operational window from 1 August to 24 November 2024, the four-class confusion-matrix diagonals are 0.58 (N), 0.57 (C), 0.63 (M), and 0.81 (X). These operational figures provide additional perspective on deployed skill, and the C-class diagonal of 0.57 is comparable to our ≥C1.0 test baselines.

Table 9. Baseline 24 h forecast skill on the seed 3970 AR-level test set ($B_{\text{th}} = 800$ G). Vision models use PNG crops; SVM uses the 29 magnetic features. Decision thresholds are selected on the validation set to maximize TSS. ROC and PR denote ROC-AUC and PR-AUC (threshold-free). For FAR, lower is better; for all other metrics, higher is better. Best values in each task block are shown in bold.

Task	Model	ACC	TPR	TNR	TSS	F1	MCC	Prec.	FAR	ROC	PR
≥C1.0	SVM	0.7771	0.6910	0.7948	0.4859	0.5141	0.4036	0.4093	0.5907	0.8166	0.5010
	ResNet-152	0.8289	0.6608	**0.8634**	0.5243	0.5685	0.4715	0.4988	0.5012	0.8436	0.6208
	ViT-B/16	0.8191	**0.7096**	0.8417	**0.5513**	0.5724	0.4774	0.4797	0.5203	**0.8601**	**0.6236**
	Swin-B	**0.8312**	0.6824	0.8618	0.5442	**0.5797**	**0.4856**	**0.5039**	**0.4961**	0.8573	0.6205
≥M1.0	SVM	0.8235	0.7873	0.8252	0.6125	0.2834	0.3140	0.1728	0.8272	0.8757	0.2981
	ResNet-152	**0.8599**	0.7983	**0.8628**	0.6611	**0.3357**	**0.3653**	**0.2125**	**0.7875**	0.9127	**0.3923**
	ViT-B/16	0.8134	**0.8693**	0.8109	0.6801	0.2924	0.3383	0.1757	0.8243	0.9159	0.3428
	Swin-B	0.8297	0.8681	0.8279	**0.6960**	0.3113	0.3562	0.1897	0.8103	**0.9178**	0.3758
≥M5.0	SVM	0.7937	0.8613	0.7931	0.6545	0.0609	0.1406	0.0316	0.9684	0.8735	0.0390
	ResNet-152	0.8817	0.8411	0.8821	0.7231	0.0995	0.1929	0.0529	0.9471	0.9250	0.0771
	ViT-B/16	0.8254	**0.9248**	0.8246	0.7494	0.0760	0.1708	0.0396	0.9604	0.9201	0.0590
	Swin-B	**0.9118**	0.8568	**0.9123**	**0.7690**	**0.1312**	**0.2317**	**0.0710**	**0.9290**	**0.9501**	**0.1153**
≥X1.0	SVM	0.6862	0.8648	0.6856	0.5504	0.0185	0.0691	0.0093	0.9907	0.8425	0.0117
	ResNet-152	0.8247	0.9896	0.8241	0.8137	0.0372	0.1240	0.0189	0.9811	0.9460	0.0322
	ViT-B/16	**0.8770**	0.9361	**0.8768**	0.8129	**0.0495**	**0.1430**	**0.0254**	**0.9746**	0.9548	**0.0499**
	Swin-B	0.8431	**1.0000**	0.8426	**0.8426**	0.0418	0.1341	0.0213	0.9787	**0.9575**	0.0434

4. Discussion

4.1. Evaluation Protocol and Class Imbalance

JW-FD adopts an **8:1:1 split at the AR level** obtained by stratified rejection sampling (seed 3970), ensuring that all magnetogram crops from a given AR appear exclusively in the training, validation, or test partition (Table 6). This protocol prevents information leakage arising from temporal autocorrelation within ARs [3,20]. The pre-registered gates require matched rare-class counts and bounded positive-frame ratios between validation and test; Table 7 reports the resulting positive-frame counts in training, validation, and test at all seven horizons. Split CSV files (`*_train.csv`, `*_val.csv`, `*_test.csv`) are distributed under `label/png/Th{threshold}/` and `label/fits/`.

Positive labels are rare for higher flare thresholds (M5.0, X1.0), producing pronounced class imbalance. At longer forecast horizons the positive fraction increases, because a qualifying flare anywhere in a longer future window yields a positive label; shorter horizons are therefore more imbalanced. Over the full release, sample-level GOES class labels at the 72 h horizon comprise 21,146 X-class, 177,099 M-class, and 594,645 C-class events, together with 1,211,079 non-flare samples. Skill oriented metrics such as TSS, F1, and MCC therefore provide a more balanced assessment than raw accuracy alone [19].

4.2. Recommended Usage

JW-FD is intended for the heliophysics and space weather ML communities. Suggested applications include:

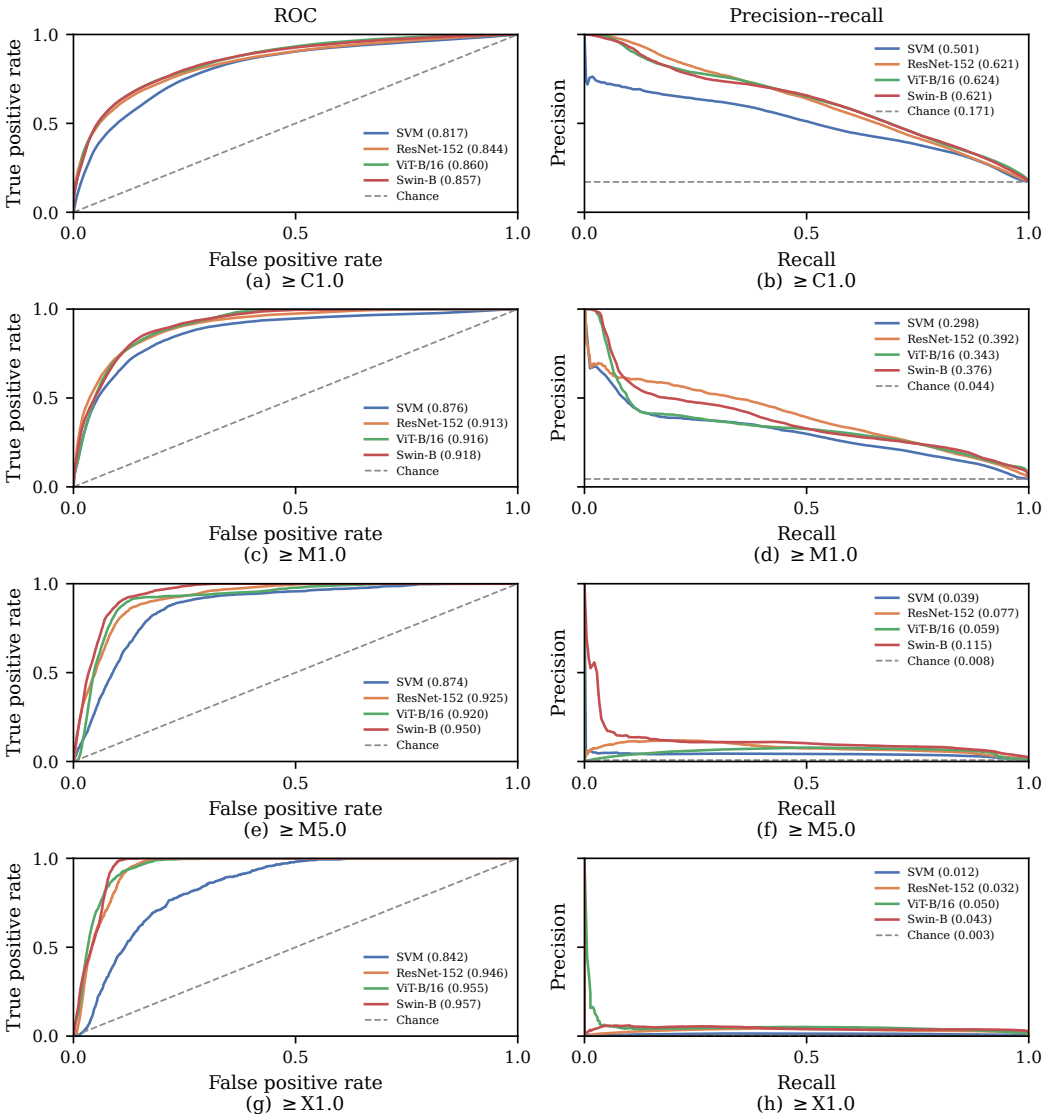


Figure 5. ROC (left) and precision–recall (right) curves on the seed 3970 AR-level test set for 24 h forecasting. Panels (a,b) ≥C1.0; (c,d) ≥M1.0; (e,f) ≥M5.0; (g,h) ≥X1.0. Chance is the diagonal on the ROC panels and the positive-class prevalence on the PR panels. Ranking remains high while precision collapses as the class becomes rarer.

- **Vision and video models**: Train CNNs, ViTs, or temporal models on PNG crops or MP4 evolution sequences, selecting B_{th} by ablation (Section 3.2) and comparing against the 24 h baselines in Section 3.3. The compromise starting clip is $B_{\text{th}} = 800$ G.
- **Parameter-based ML**: Apply classical classifiers or gradient boosted trees to the 29 dimensional magnetic parameter vectors, comparing FITS derived (Gauss scale) and PNG derived (intensity scale) branches.
- **Multimodal fusion**: Combine PNG/MP4 inputs with magnetic parameters and labels from the same indexed record [6,7].
- **Multi horizon benchmarking**: Evaluate models across seven forecast horizons and four GOES thresholds using the precomputed label columns.

The approximately 2.00 million magnetogram frames originate from 3,083 active regions and are strongly autocorrelated in time within each region; they should not be treated as two million independent physical samples. The 24 h baselines in Section 3.3 make this concrete for snapshot models. Training retained every active region but temporally subsampled frames, and on the seed 3970 AR-level test set an SVM on the 29 magnetic

features is only modestly below ResNet-152, ViT-B/16, and Swin-B. JW-FD therefore supports parameter-based ML and single-frame vision on a similar footing; it is not a vision-only corpus, and the image count should not be read as an ImageNet-scale advantage. From $\geq$C1.0 to $\geq$M1.0 the TSS remains of order 0.49–0.70, whereas precision and F1 fall and the false-alarm rate exceeds 0.78, so a single-frame score is a poor operational alert for rarer classes.

Sequence and video models were not trained here, but they are a natural match to the same statistics: the within-AR autocorrelation that invalidates i.i.d. frame counts is the temporal structure those models are meant to use, and the release already provides tracked 12 min crops, seven forecast horizons, and timestamp-aligned PNG, CSV, and MP4 records. The MP4 files are a 60 fps playback of those native frames, not a high-cadence video stream, so video architectures should be treated as temporal models on the HMI cadence rather than as conventional video recognizers. In all of these settings the independent unit remains the active region.

Analyses that assume independent and identically distributed observations, that require cycle-to-cycle hold-out, that treat MP4 clips as independent videos, or that target the rarest X-class events with fine-grained uncertainty from frame counts alone are less well supported by the nominal sample size. JW-FD is particularly suitable for AR-level generalization studies, multi-horizon and multi-threshold benchmarking, snapshot parameter or vision baselines, and sequence or multimodal experiments that exploit the aligned records. For rare-class tasks, we recommend reporting skill scores that are robust to imbalance (TSS, F1, MCC, together with precision or FAR), quantifying uncertainty at the active-region level, and, where appropriate, constructing chronological splits as noted in Section 4.3.

Beyond offline forecasting benchmarks, JW-FD can also support AI driven solar observing systems that need flare aware context or multimodal solar representations [26–28].

To reproduce or extend the dataset, users can clone the open-source pipeline at https://github.com/Xiaoxuan-1/JW-FD and set `START_YEAR=2011`, `END_YEAR=2025` in `config.py`. We recommend using the distributed split files so that evaluations remain comparable across publications. Those files implement the seed 3970 assignment of Table 6. The compromise PNG working point is $B_{\mathrm{th}} = 800$ G; all ten clips remain available for re-ablation.

4.3. Limitations

JW-FD is restricted to HMI line-of-sight magnetograms and does not ingest vector-field products such as SHARP [5]. No geometric projection correction is applied: the released 600×600 crops remain in the observer plane, and B_{LoS} is not converted to an approximate radial field. Foreshortening and the viewing-angle dependence of the LoS component therefore remain in both the images and the 29 derived features. The $|\mathrm{longitude}| \leq 60^{\circ}$ filter introduces a disk-center bias and excludes high-latitude and near-limb active regions; it mitigates the most severe limb effects but is not a substitute for heliographic or cylindrical equal-area remapping. AR identification depends on NOAA numbering and SRS daily positions, which can lag rapid flux emergence. The native HMI cadence of approximately 12 min may also undersample rapidly evolving pre-flare configurations.

In addition, the recommended 8:1:1 split is a stratified rejection sample of active regions across the full 2011–2025 interval. It therefore tests generalization to unseen active regions under similar solar-cycle conditions, and it should not be interpreted as demonstrating generalization across solar cycles or to independent future epochs. The present coverage spans the declining phase of Cycle 24 through the rising and maximum phase of Cycle 25

and cannot by itself establish cycle-to-cycle generalization. Users interested in temporal generalization may construct chronological evaluation splits from the released timestamps.

5. Conclusions

JW-FD is a 15-year multimodal dataset for solar-flare forecasting. It comprises 3,083 independent NOAA active regions and 2.00 million coregistered line-of-sight magnetogram crops spanning 1 January 2011 through 31 December 2025, released as aligned FITS, PNG, CSV, and MP4 products. PNG images, 29-dimensional magnetic features, and flare labels are provided at ten saturation thresholds from 200 to 2000 G, together with an AR-level 8:1:1 split that limits temporal leakage and, by stratified rejection sampling (seed 3970), matches rare-class positives between validation and test at all seven forecast horizons. The compromise PNG working point is $B_{\mathrm{th}} = 800$ G. JW-FD therefore supplies a large, leakage-controlled benchmark that links HMI magnetic observations with NOAA active-region and flare information across multiple prediction timescales and flare thresholds.

The dataset is intended as common, reusable infrastructure for traditional machine learning, deep learning, and multimodal methods. The baseline experiments show that the released records can support both parameter-based and vision-based models. Subsequent work can build on these records by exploiting the evolution videos, fusing image and parameter branches, incorporating vector magnetic-field products, and constructing chronological evaluation splits when cycle-to-cycle generalization is the scientific goal. We regard the present release as a foundation for that future investigation and optimization.

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Data Availability Statement: The JW-FD construction pipeline is openly available at https://github.com/Xiaoxuan-1/JW-FD. Zenodo version v1 (https://doi.org/10.5281/zenodo.21672850) is a representative subset of about 10% of an earlier release. Version v2 (https://doi.org/10.57760/sciencedb.019mg), the release corresponding to this paper, is deposited in the Science Data Bank (ScienceDB). Relative to the published JW-FD v1, this version restores the many flares that the earlier catalog omitted by retaining only events marked with a plus sign, and rebuilds the training, validation, and test partitions so that their sample distributions are balanced. The number of frames with an M1.0 or stronger flare within the next 24 h increases from 74,000 in v1 to 103,626 in v2. September 2011 magnetograms that an earlier build did not read are included in this count.

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